A Lightweight Diffusion–Transformer Framework with Knowledge Distillation for Uncertainty-Aware Diabetic Retinopathy Detection

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Problem Statement

Real-world diabetic retinopathy (DR) screening often relies on edge and IoT devices operating under poor imaging conditions, resulting in low-quality fundus images.

Core problems:

- Fundus images captured on edge devices often suffer from low quality.
- Accurate DR models are large and cannot run on edge devices.
- Lightweight models lose important diagnostic information.
- Knowledge distillation helps, but struggles under low-quality images and teacher–student mismatch.

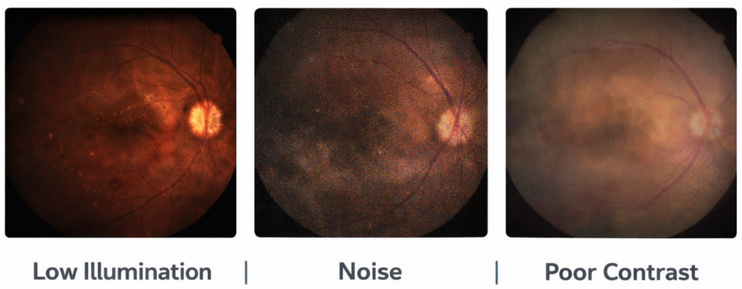

Figure 1. Low-quality fundus images in real-world diabetic retinopathy screening.

Therefore, the core problem is designing a lightweight framework that jointly addresses image degradation, efficient knowledge transfer, and reliable uncertainty estimation for DR detection on edge devices.

Motivation

Diabetic Retinopathy is a major cause of preventable blindness, yet many patients do not receive timely screening due to limited access to specialists and medical infrastructure. In many regions, screening relies on low-cost devices that produce imperfect images and cannot support complex diagnostic models. This research is motivated by enabling early, safe, and practical DR screening, especially in resource-constrained settings.

Objectives

- To develop a lightweight diffusion–transformer framework for diabetic retinopathy detection under low-quality imaging conditions.
- To enhance fundus image quality using diffusion-based restoration while preserving critical diagnostic features.
- To improve efficiency and performance through knowledge distillation with adapter-based student models suitable for edge deployment.
- To incorporate uncertainty estimation for reliable decision-making in real-world scenarios.

Literature Review

Year	Authors	Method Focus	Dataset(s)	Limitation / Gap
2023	Wang et al.[4]	Adapter-based ViT for lesion segmentation	IDRiD, DDR	No low-light enhancement or uncertainty modeling
2025	Chen et al.[1]	Diffusion-based DR classification	APTOS, Messidor	Computationally heavy; not edge-friendly
2024	Dutt et al.[7]	Parameter-efficient fine-tuning (PEFT)	Multiple datasets	Not DR-specific; lacks uncertainty analysis
2024	Akram et al.[5]	Uncertainty-aware DR detection (Bayesian DL)	APTOS, DDR	Large backbone; high inference cost
2025	Dejene et al.[2]	Knowledge distillation for lightweight DR models	APTOS 2019	Sensitive to image quality; no enhancement
2025	Zhang et al.[3]	Diffusion-based augmentation for DR grading	DDR, EyePACS	No focus on deployment or uncertainty
2025	Chakour et al.[6]	Mobile/edge DR screening system	APTOS, EyePACS	Limited robustness under poor imaging

Research Question?

- How can low-light fundus images be enhanced to improve diabetic retinopathy detection performance?
- Can knowledge distillation enable a lightweight student model to achieve performance comparable to a high-capacity teacher model?
- How can uncertainty estimation be incorporated to improve the reliability of DR detection systems?
- Is the proposed framework suitable for deployment on resource-constrained edge devices?

Proposed Methodology

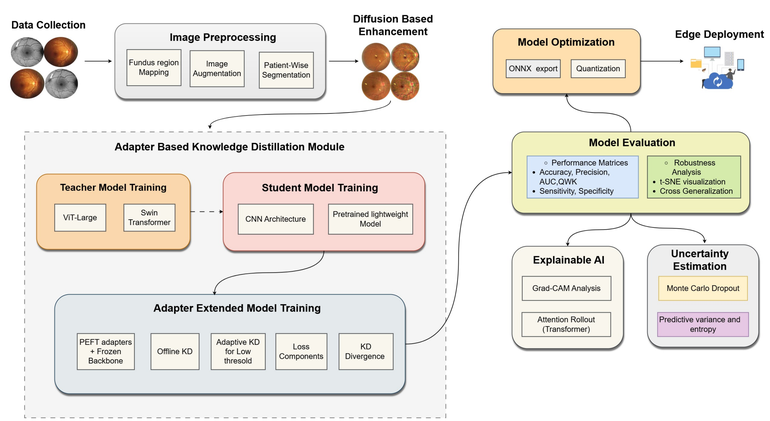

Figure 2. Overview of the proposed Methodology of this study

Proposed Workflow Framework

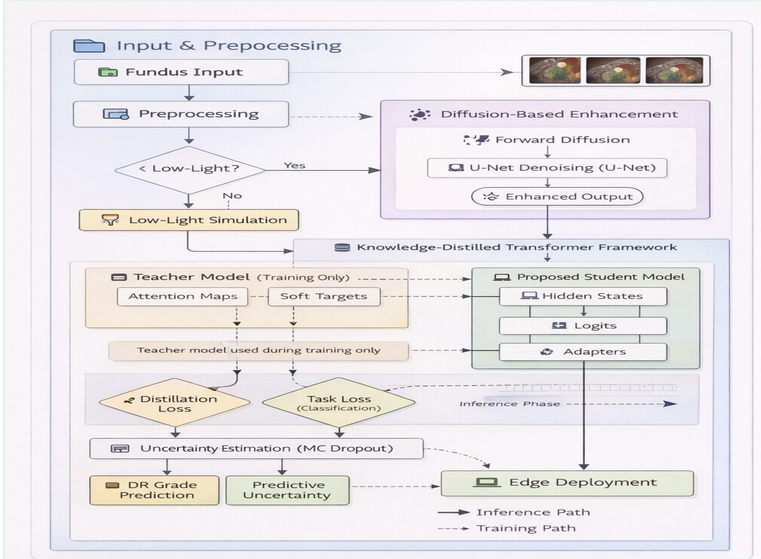

Figure 3. Overview of the proposed workflow framework.

Expected Outcomes and Conclusion

- Improved diabetic retinopathy detection performance on low-quality fundus images with a lightweight and edge-friendly model.
- Reliable predictions with quantified uncertainty to support safe clinical decision-making and referral.

References

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